

L05: Learning in games

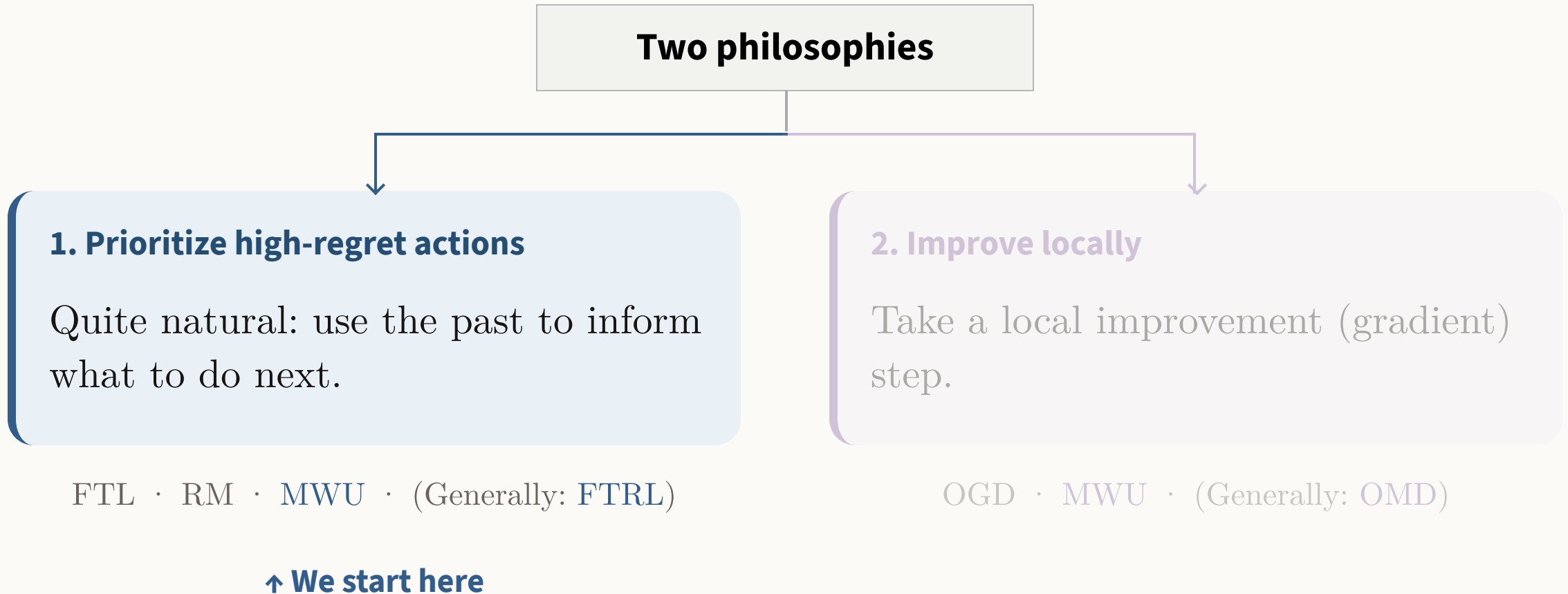
Algorithms



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– How do we construct a regret minimizer?



– Let's track regrets

At every time t , the *instantaneous regret* accumulated against each action is

$$\delta_a^{(t)} := g_a^{(t)} - \langle \mathbf{g}^{(t)}, \mathbf{x}^{(t)} \rangle$$

In vector form:

$$\boldsymbol{\delta}^{(t)} := \mathbf{g}^{(t)} - \langle \mathbf{g}^{(t)}, \mathbf{x}^{(t)} \rangle \mathbf{1}$$

Hence, the vector of regrets against each action is

$$\mathbf{r}^{(t)} := \sum_{\tau=1}^t \boldsymbol{\delta}^{(\tau)}, \quad \text{and} \quad \text{Reg}^{(T)} = \max_a r_a^{(T)}$$

Follow-the-leader

Idea #1: Pick the action with *highest regret*

– Follow-the-leader

Play the action we most wish we had used in the past.

All mass on the action with highest regret

$$\mathbf{x}^{(t+1)} = \mathbf{e}_{\arg \max_a r_a^{(t)}}$$

Equivalently, maximize cumulative utility:

$$\arg \max_a r_a^{(t)} = \arg \max_a \sum_{\tau=1}^t g_a^{(\tau)}$$

What could go wrong?

– Follow-the-leader

t	1	2	3	4	5	6
FTL plays	a_1	a_2	a_1	a_2	a_1	a_2
$g^{(t)}$	(0, 0.5)	(1, 0)	(0, 1)	(1, 0)	(0, 1)	(1, 0)
FTL total	0	0	0	0	0	0
Best fixed total	0.5	1	1.5	2	2.5	3

Regret: 3

The leader can always be one round behind: $\text{Reg}^{(T)}$ could be $\Omega(T)$ in general)

Intuitive issue: the algorithm “*jumps too much*”. It is “not smooth”...

– Not all hope is lost!

Interesting result: If both players in a *two-player zero-sum game* use FTL, we obtain *fictitious play*.

Julia Robinson (1951)

In every finite two-player zero-sum game, the *average strategies* approach the set of Nash equilibria.

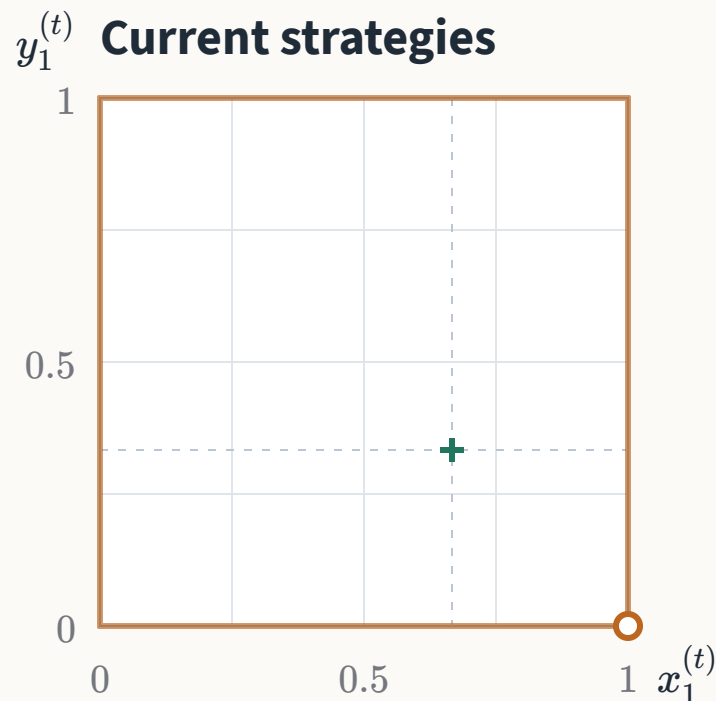
(*An iterative method of solving a game*, 1950.)

Convergence can be very slow. But this shows that if everyone is learning, FTL has sublinear regret...

- Fictitious play in a zero-sum game

Current play keeps switching; averages approach Nash

Each player best responds to the opponent's average strategy so far.



$$U = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

$$u_1 = \mathbf{x}^\top U \mathbf{y}$$

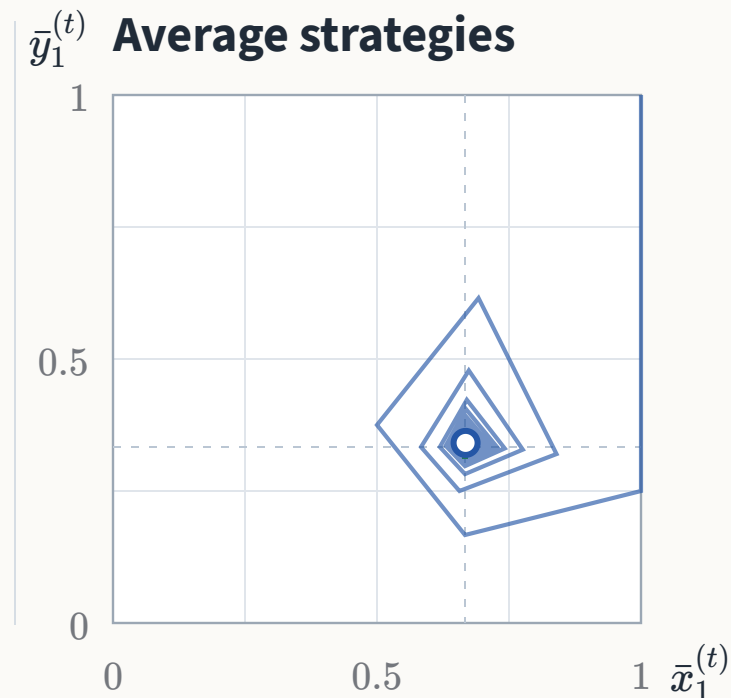
$$u_2 = -u_1$$

Pure best responses

$$(x_1, y_1) = (1, 0)$$

Equilibrium gap

1.000



+ Nash equilibrium

$$(x_1^*, y_1^*) = \left(\frac{2}{3}, \frac{1}{3}\right)$$

Running average

$$(0.668, 0.341)$$

Equilibrium gap

0.009

Simultaneous updates · Ties: choose the first action

– How do we prove the no-regret property?

Two approaches

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graph TD; A[Two approaches] --> B[1. Against arbitrary adversaries]; A --> C[2. Within a more benign environment];
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1. Against arbitrary adversaries

Seek to prove that the property holds no matter what the environment does

Pros: no need to think about the world, robust

Cons: higher bar, more conservative/pessimistic

↑ **Online learning approach**

2. Within a more benign environment

If everyone is learning, the dynamics are somewhat smooth. Full adversarialness is not unleashed

Pros: better results *are* possible (eg, *fictitious play*)

Cons: analysis is harder, makes assumptions about the environment

Regret matching

Idea #2: Pick actions *proportional** to their regret

* Wait a second: regret can be negative. So let's first threshold at zero...

– Regret matching

Second simplest idea: allocate mass proportional to regret

Use the positive part: $[r_a]^+ := \max\{r_a, 0\}$.

Play in proportion to positive regret

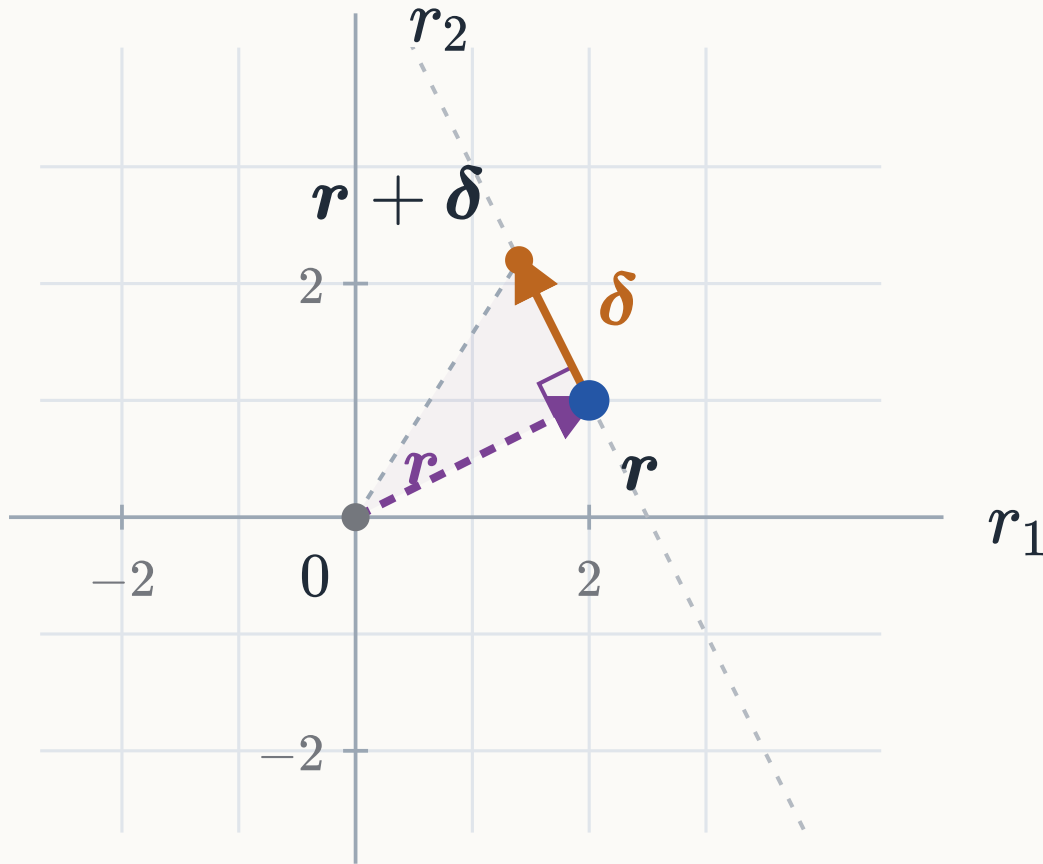
$$\mathbf{x}^{(t+1)} = \frac{[\mathbf{r}^{(t)}]^+}{\|[\mathbf{r}^{(t)}]^+\|_1}$$

If all regrets are nonpositive, use any distribution.

$$\mathbf{r} = (2, -1, 1) \quad \mathbf{x} = ?$$

$$\mathbf{x} = (2/3, 0, 1/3)$$

- The geometry of regret matching



First, a simplifying assumption

Suppose the current $\mathbf{r} \geq \mathbf{0}$.

The new regret increment

$$\delta = \mathbf{g} - \langle \mathbf{g}, \mathbf{x} \rangle \mathbf{1}$$

Just Pythagoras

$\langle \mathbf{x}, \delta \rangle = 0$ for every strategy.

RM chooses $\mathbf{x} \propto \mathbf{r}$.

$$\langle \mathbf{r}, \delta \rangle = 0$$

$$\|\mathbf{r} + \delta\|_2^2 = \|\mathbf{r}\|_2^2 + \|\delta\|_2^2$$

$$\mathbf{x} = (0.67, 0.33)$$

– The general case: regrets can be negative

Now drop the assumption. Track the squared norm of the positive part: $\|[\mathbf{r}]^+\|_2^2$

1. Orthogonality: $\langle [\mathbf{r}]^+, \boldsymbol{\delta} \rangle = \|[\mathbf{r}]^+\|_1 \langle \mathbf{x}, \boldsymbol{\delta} \rangle = 0$.

2. Inequality + Pythagoras:

$$\begin{aligned}\|[\mathbf{r} + \boldsymbol{\delta}]^+\|_2^2 &\leq \|[\mathbf{r}]^+ + \boldsymbol{\delta}\|_2^2 \\ &= \|[\mathbf{r}]^+\|_2^2 + \underbrace{2\langle [\mathbf{r}]^+, \boldsymbol{\delta} \rangle}_{=0} + \|\boldsymbol{\delta}\|_2^2\end{aligned}$$

Hence,

$$\|[\mathbf{r}^{(T)}]^+\|_2^2 \leq \sum_{t=1}^T \|\boldsymbol{\delta}^{(t)}\|_2^2$$

Right-hand side scales linearly in T , which implies regret bounded by $\approx \sqrt{T} \dots$

– Regret matching: regret bound

Assuming $\|\mathbf{g}^{(t)}\|_\infty \leq 1$ at all times t ,

$$\begin{aligned}\|\boldsymbol{\delta}^{(t)}\|_2 &= \|\mathbf{g}^{(t)} - \langle \mathbf{g}^{(t)}, \mathbf{x}^{(t)} \rangle \mathbf{1}\|_2 \\ &\leq \|\mathbf{g}^{(t)}\|_2 + |\langle \mathbf{g}^{(t)}, \mathbf{x}^{(t)} \rangle| \cdot \|\mathbf{1}\|_2 \\ &\leq \sqrt{|A|} + \sqrt{|A|} = 2\sqrt{|A|}\end{aligned}$$

Hence

$$\text{Reg}^{(T)} = \max_a r_a^{(T)} \leq \sqrt{\|\mathbf{r}^{(T)}\|_2^2} \leq \sqrt{T \cdot 4|A|} = 2\sqrt{T|A|}.$$

Great news! Sublinear regret with no stepsize to tune

$$\boxed{\text{Reg}^{(T)} \leq 2\sqrt{T|A|}}$$

Against *any* sequence of bounded utility vectors. No stepsize to tune!

Multiplicative weights update

Idea #3: Avoid instability by *smoothing* the maximum

– Smoothing follow-the-leader

$$x_a = \frac{\exp(\eta r_a)}{\sum_b \exp(\eta r_b)}$$

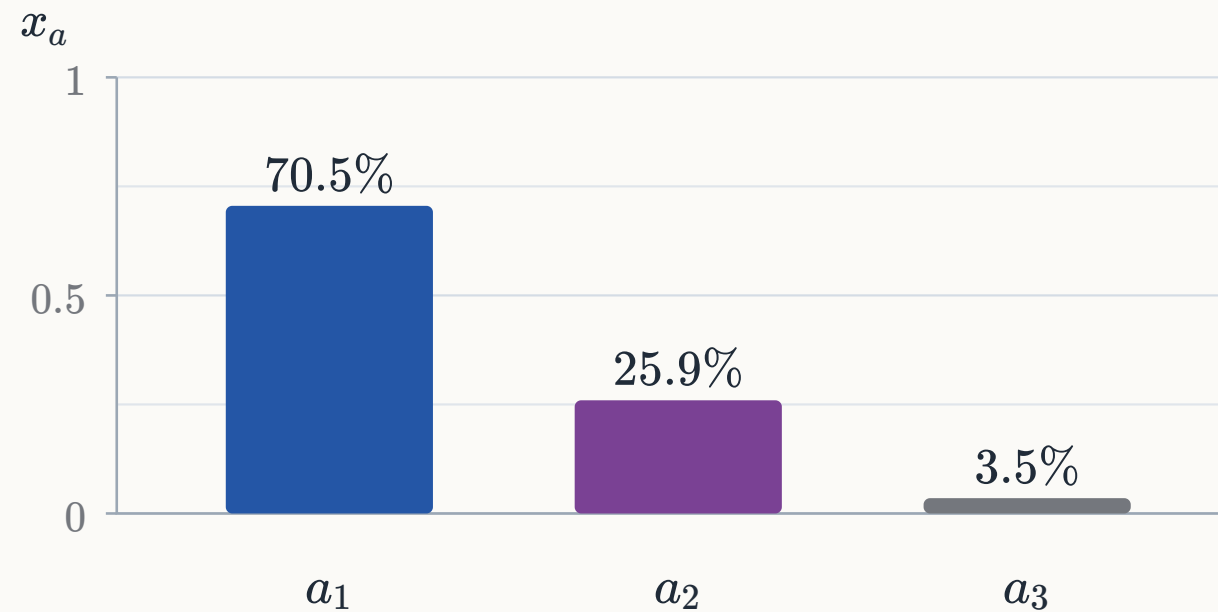
$\eta \downarrow 0$: uniform

$\eta \uparrow \infty$: leader

The stepsize controls sensitivity

Small stepsizes temper the influence of past scores. Large stepsizes approach the hard leader.

Cumulative regret $\mathbf{r} = (2, 1, -1)$



Small η : nearly uniform. Large η : nearly follow-the-leader.

– MWU: regret bound

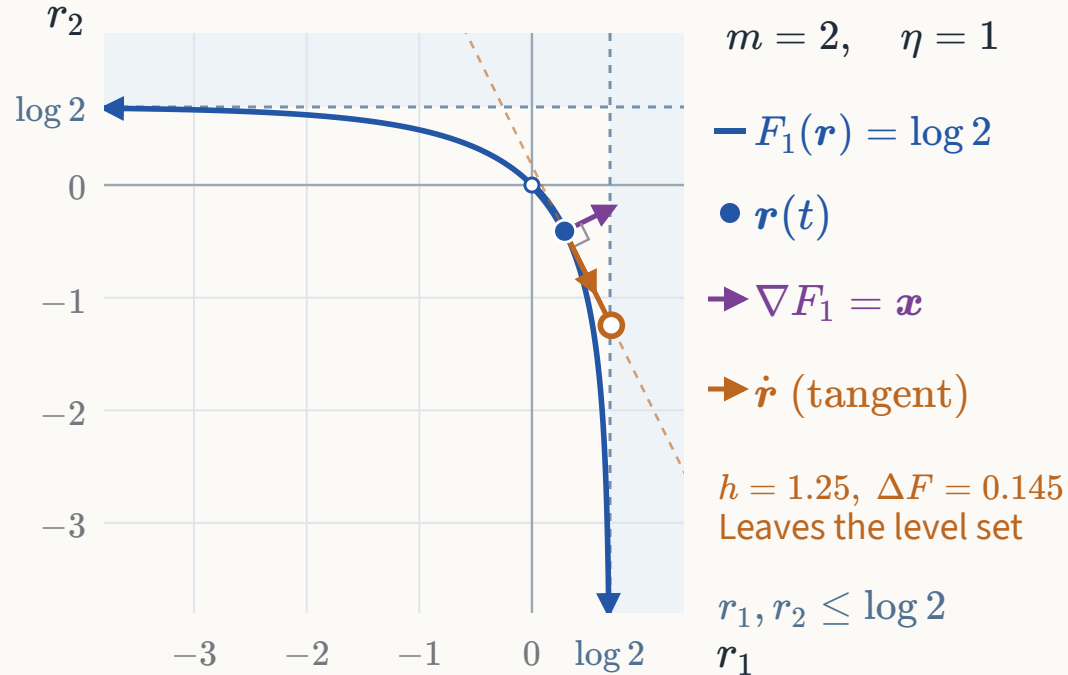
$$\text{Reg}^{(T)} \leq \frac{\log |A|}{\eta} + \eta \sum_{t=1}^T \|\mathbf{g}^{(t)}\|_{\infty}^2$$

Hence, by setting $\eta = \sqrt{\log |A|/T}$,

$$\text{Reg}^{(T)} \leq 2\sqrt{T \log |A|}$$

Compared to RM: Logarithmic dependence on the number of actions, but one hyperparameter (η)

– Continuous-time MWU



Softmax is the gradient of log-sum-exp

$$F_\eta(\mathbf{r}) := \frac{1}{\eta} \log \sum_{a=1}^m \exp(\eta r_a), \quad m = |A|$$

$$(\nabla F_\eta(\mathbf{r}))_a = \frac{\exp(\eta r_a)}{\sum_b \exp(\eta r_b)} = x_a$$

Infinitesimal updates stay on one level set

$$\dot{\mathbf{r}} = \mathbf{g} - \langle \mathbf{g}, \mathbf{x} \rangle \mathbf{1}$$

$$\frac{d}{dt} F_\eta(\mathbf{r}(t)) = \langle \mathbf{x}, \dot{\mathbf{r}} \rangle = 0$$

$$\max_a r_a(t) \leq F_\eta(\mathbf{r}(t)) = F_\eta(\mathbf{0}) = \frac{\log m}{\eta}$$

Discrete time: the same potential, plus discretization error

$$\text{Reg}^{(T)} \leq \frac{\log m}{\eta} + \eta \sum_{t=1}^T \|\mathbf{g}^{(t)}\|_\infty^2$$

- How do we construct a regret minimizer?



- Online projected gradient ascent

The second philosophy: take a utility-gradient step (and project back)

Projected gradient ascent:

$$\mathbf{x}^{(t+1)} := \Pi_{\mathcal{X}} \left(\mathbf{x}^{(t)} + \eta \mathbf{g}^{(t)} \right)$$

where

$$\Pi_{\mathcal{X}}(\mathbf{z}) := \arg \min_{\mathbf{x} \in \mathcal{X}} \|\mathbf{x} - \mathbf{z}\|_2^2$$

– OGD: regret bound

$$\text{Reg}^{(T)} \leq \frac{D^2}{2\eta} + \frac{\eta}{2} \sum_{t=1}^T \|\mathbf{g}^{(t)}\|_2^2$$

where $D = \text{diam}_2(\mathcal{X})$. Proof idea: again a telescopic argument (we will see it in the next slide).

On the simplex

$$D \leq \sqrt{2}, \quad \eta = \sqrt{\frac{2}{T|A|}} \implies \text{Reg}^{(T)} \leq \sqrt{2T|A|}$$

– The OGD proof

2. Now, the term on the right-hand side satisfies

$$\|\hat{\mathbf{x}} - (\mathbf{x}^{(t)} + \eta \mathbf{g}^{(t)})\|_2^2 = \|\hat{\mathbf{x}} - \mathbf{x}^{(t)}\|_2^2 - 2\eta \langle \mathbf{g}^{(t)}, \hat{\mathbf{x}} - \mathbf{x}^{(t)} \rangle + \eta^2 \|\mathbf{g}^{(t)}\|_2^2.$$

3. Rearranging:

$$\langle \mathbf{g}^{(t)}, \hat{\mathbf{x}} - \mathbf{x}^{(t)} \rangle \leq \frac{\|\hat{\mathbf{x}} - \mathbf{x}^{(t)}\|_2^2 - \|\hat{\mathbf{x}} - \mathbf{x}^{(t+1)}\|_2^2}{2\eta} + \frac{\eta}{2} \|\mathbf{g}^{(t)}\|_2^2$$

4. Sum over time. The distances telescope.

⇒ OGD regret bound

$$\text{Reg}^{(T)} \leq \frac{D^2}{2\eta} + \frac{\eta}{2} \sum_{t=1}^T \|\mathbf{g}^{(t)}\|_2^2$$

where $D = \text{diam}_2(\mathcal{X})$

– The guarantees

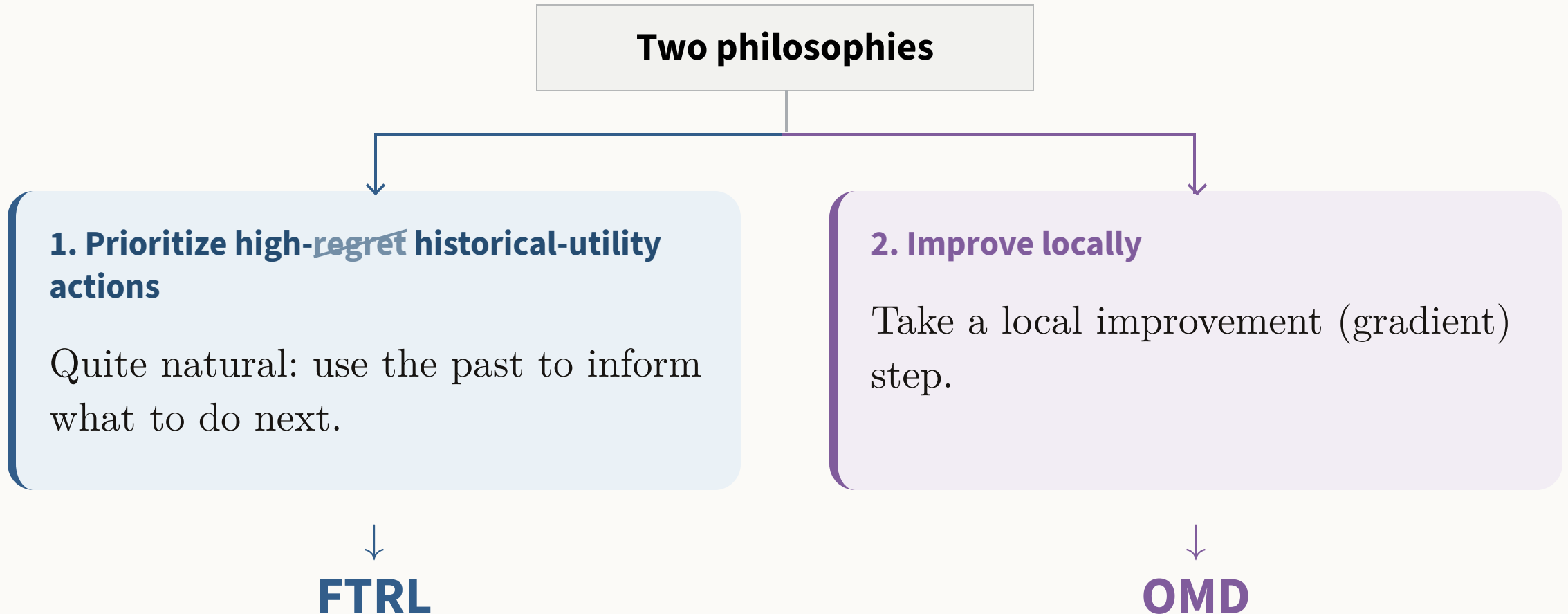
(Assuming $\|\mathbf{g}^{(t)}\|_\infty \leq 1$)

Algorithm	Stepsize	Regret bound
FTL	—	$T/2$ on our sequence
RM	None ✨	$2\sqrt{T A }$
MWU	$\sqrt{\log A /T}$	$2\sqrt{T \log A }$
OGD	$\sqrt{2/(T A)}$	$\sqrt{2T A }$

Beyond normal-form games

The *big guns*: FTRL and OMD

- Two philosophies, two frameworks



Convex strategy sets, concave utilities. Use $\mathbf{g}^{(t)} = \nabla u^{(t)}(\mathbf{x}^{(t)})$.

– Both algorithms work in a tunable geometry

Tune FTRL and OMD to the strategy set

Choose a strongly convex regularizer ψ , also called a *distance-generating function*.

Strongly curved upward

For twice-differentiable ψ , in Euclidean geometry:

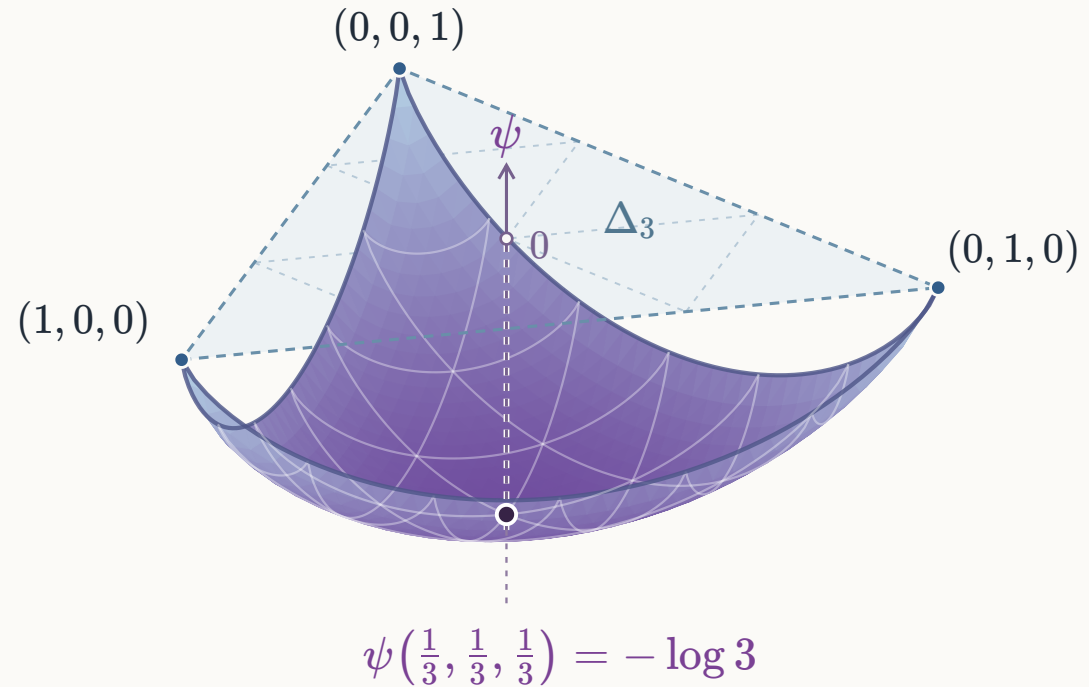
$$\nabla^2 \psi(\mathbf{x}) \succeq I$$

A safe choice is always the *squared Euclidean norm*, though it might not be the optimal choice depending on the geometry of the strategy set...

Negative entropy on the simplex

$$\psi(\mathbf{x}) = \sum_{a=1}^3 x_a \log x_a$$

$$x_3 = 1 - x_1 - x_2$$



- Follow-the-regularized-leader

Stabilize the leader with regularization

Balance cumulative utility against a fixed regularizer.

$$\mathbf{x}^{(t+1)} = \arg \max_{\mathbf{x} \in \mathcal{X}} \left\{ \underbrace{\left\langle \sum_{\tau=1}^t \mathbf{g}^{(\tau)}, \mathbf{x} \right\rangle}_{\text{cumulative utility}} - \frac{1}{\eta} \underbrace{\psi(\mathbf{x})}_{\text{fixed regularizer}} \right\}$$

Small η

Regularization dominates.

No movement (stay at minimizer of ψ).

Large η

Cumulative utility dominates.

Too much movement ($\equiv$ follow-the-leader).

- MWU = FTRL with Entropy

$$\mathbf{x}^{(t+1)} = \arg \max_{\mathbf{x} \in \mathcal{X}} \left\{ \left\langle \sum_{\tau=1}^t \mathbf{g}^{(\tau)}, \mathbf{x} \right\rangle - \frac{1}{\eta} \psi(\mathbf{x}) \right\}, \quad \boxed{\psi(\mathbf{x}) = \sum_a x_a \log x_a}.$$

What is the solution to the optimization problem?

Use Lagrange multipliers on the plane of the simplex (ignore nonnegativity constraints):

$$\begin{aligned} \left(\sum_{\tau=1}^t \mathbf{g}^{(\tau)} \right) - \frac{1}{\eta} (\mathbf{1} + \log \mathbf{x}) &= \lambda \mathbf{1} \\ \implies \log \mathbf{x} &= c \mathbf{1} + \eta \sum_{\tau=1}^t \mathbf{g}^{(\tau)} \implies \mathbf{x} \propto \exp \left\{ \eta \sum_{\tau=1}^t \mathbf{g}^{(\tau)} \right\} \end{aligned}$$

which is MWU.

- FTRL: Regret Bound

FTRL guarantees

$$\text{Reg}^{(T)} \leq \frac{B}{\eta} + \eta \sum_{t=1}^T \|\mathbf{g}^{(t)}\|_*^2$$

Where

$$B_{\text{FTRL}} = \max_{\mathbf{x}} \psi - \min_{\mathbf{x}} \psi$$

Hence, assuming

$$\|\mathbf{g}^{(t)}\|_* \leq G, \quad \eta = \sqrt{\frac{B}{TG^2}} \implies \text{Reg}^{(T)} \leq 2G\sqrt{BT}$$

- Online mirror descent

Choose a geometry for each local step

Balance the latest utility feedback against moving from the current strategy.

$$\mathbf{x}^{(t+1)} = \arg \max_{\mathbf{x} \in \mathcal{X}} \left\{ \underbrace{\langle \mathbf{g}^{(t)}, \mathbf{x} \rangle}_{\text{latest feedback}} - \frac{1}{\eta} \underbrace{D_{\psi}(\mathbf{x} \parallel \mathbf{x}^{(t)})}_{\text{movement penalty}} \right\}$$

where

$$D_{\psi}(\mathbf{z} \parallel \mathbf{x}) := \psi(\mathbf{z}) - \psi(\mathbf{x}) - \langle \nabla \psi(\mathbf{x}), \mathbf{z} - \mathbf{x} \rangle$$

Small η

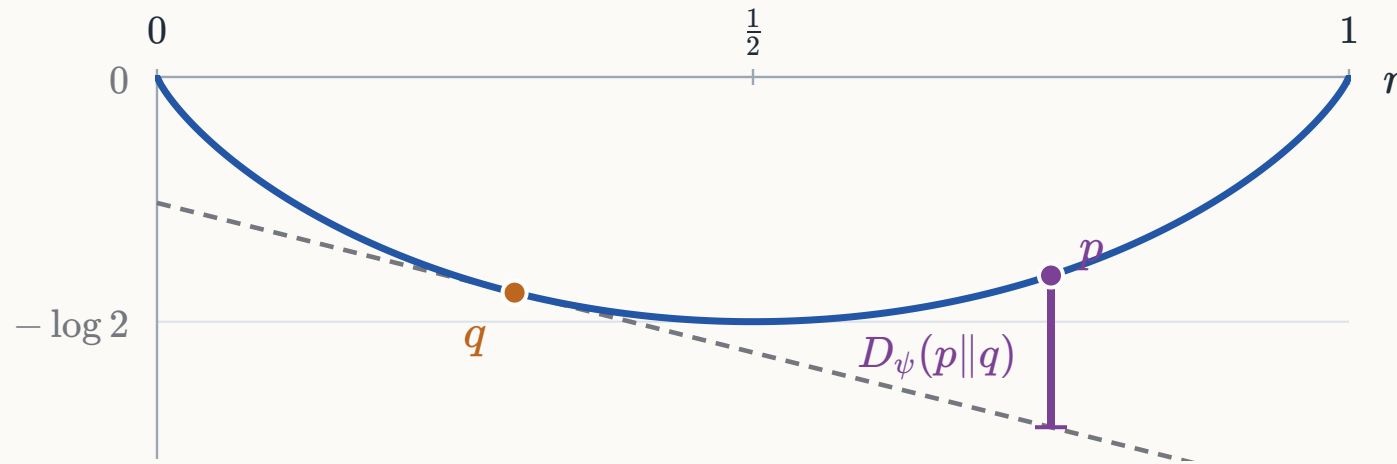
Movement penalty dominates.
Stay near $\mathbf{x}^{(t)}$.

Large η

Latest feedback dominates.
Maximize the current linear utility.

- Negative entropy and its tangent

$$\psi(\mathbf{x}) = \sum_a x_a \log x_a, \quad \mathbf{x} = (p, 1 - p), \quad \mathbf{q} = (q, 1 - q)$$



Drag the points along the curve

— Negative entropy

$$\psi(r) = r \log r + (1 - r) \log(1 - r)$$

--- Tangent at current q

| Vertical gap = KL divergence

$$D_\psi(p||q) = \text{KL}((p, 1 - p) || (q, 1 - q))$$

$$D_\psi(p||q) = 0.43$$

Bregman divergence is the height above the tangent

$$D_\psi(\mathbf{x}||\mathbf{q}) = \sum_a x_a \log(x_a/q_a) = D_{\text{KL}}(\mathbf{x}||\mathbf{q}).$$

- The geometry chooses the algorithm

Regularizer ψ	Divergence	Algorithm
$\sum_a x_a \log x_a$	D_{KL}	MWU FTRL and OMD
$\frac{1}{2} \ \mathbf{x}\ _2^2$	$\frac{1}{2} \ \mathbf{z} - \mathbf{x}\ _2^2$	OGD OMD

MWU and OGD are instances of general principles

Entropy gives MWU. Euclidean mirror descent gives OGD; quadratic FTRL can differ under constraints.

- OMD: Regret Bound

OMD guarantees

$$\text{Reg}^{(T)} \leq \frac{B}{\eta} + \eta \sum_{t=1}^T \|\mathbf{g}^{(t)}\|_*^2$$

Where

$$B_{\text{OMD}} = \max_{\mathbf{x} \in \mathcal{X}} D_{\psi}(\mathbf{x} \| \mathbf{x}^{(1)})$$

Hence, assuming

$$\|\mathbf{g}^{(t)}\|_* \leq G, \quad \eta = \sqrt{\frac{B}{TG^2}} \implies \text{Reg}^{(T)} \leq 2G\sqrt{BT}$$

- FTRL and OMD can overlap

For example: OMD with negative entropy

$$\psi(\mathbf{x}) = \sum_a x_a \log x_a, \quad D_\psi(\mathbf{x} \parallel \mathbf{x}^{(t)}) = \sum_a x_a \log(x_a / x_a^{(t)}).$$

gives rise to

$$\log x_a^{(t+1)} = \log x_a^{(t)} + \eta g_a^{(t)} + c.$$

And hence

$$x_a^{(t+1)} = \frac{x_a^{(t)} \exp(\eta g_a^{(t)})}{\sum_b x_b^{(t)} \exp(\eta g_b^{(t)})}.$$

Unrolling gives the same update as **entropic FTRL**:

$$x_a^{(t+1)} \propto \exp\left(\eta \sum_{\tau=1}^t g_a^{(\tau)}\right).$$

Appendix

Faster rates & Iterate convergence

– Optimism: add a momentum term

Nonstationary need not mean adversarial

Use the latest gradient to anticipate the next one.

In reflected OMD, replace the feedback by

$$\hat{\mathbf{g}}^{(t)} = \mathbf{g}^{(t)} + \underbrace{(\mathbf{g}^{(t)} - \mathbf{g}^{(t-1)})}_{\text{momentum correction}} = 2\mathbf{g}^{(t)} - \mathbf{g}^{(t-1)}$$

In zero-sum self-play, this can damp cycling: a negative-momentum effect.

Optimistic FTRL / OMD in zero-sum self-play

$\gamma(\bar{\mathbf{x}}, \bar{\mathbf{y}}) = O(1/T)$ with suitable variants and a sufficiently small fixed stepsize.

– Optimism in both frameworks

Add a momentum correction. Set $\mathbf{g}^{(0)} = \mathbf{0}$.

FTRL: count the latest gradient once more

$$\mathbf{x}^{(t+1)} = \arg \max_{\mathbf{x} \in \mathcal{X}} \left\{ \left\langle \sum_{\tau=1}^t \mathbf{g}^{(\tau)} + \mathbf{g}^{(t)}, \mathbf{x} \right\rangle - \frac{1}{\eta} \psi(\mathbf{x}) \right\}$$

Reflected OMD: add the change in gradients

$$\mathbf{x}^{(t+1)} = \arg \max_{\mathbf{x} \in \mathcal{X}} \left\{ \left\langle \mathbf{g}^{(t)} + (\mathbf{g}^{(t)} - \mathbf{g}^{(t-1)}), \mathbf{x} \right\rangle - \frac{1}{\eta} D_{\psi}(\mathbf{x} \parallel \mathbf{x}^{(t)}) \right\}$$

The frameworks can differ under constraints. Further variants: [Supplementary S04](#).

– Faster regret in general games

Method / setting	Each player's regret	CCE error
Optimistic FTRL / OMD Syrnganis et al. (2015)	$O(T^{1/4})$	$O(T^{-3/4})$
Optimistic MWU · normal-form games Daskalakis, Fishelson, Golowich (2021)	$O(\log^4 T)$	$O(\log^4 T / T)$
Specialized optimistic FTRL · convex games Farina et al. (2022) · lifted space, log regularizer	$O(\log T)$	$O(\log T / T)$

Cautious optimism: $O_T(\log T)$ — [Soleymani, Piliouras & Farina \(2025\)](#).

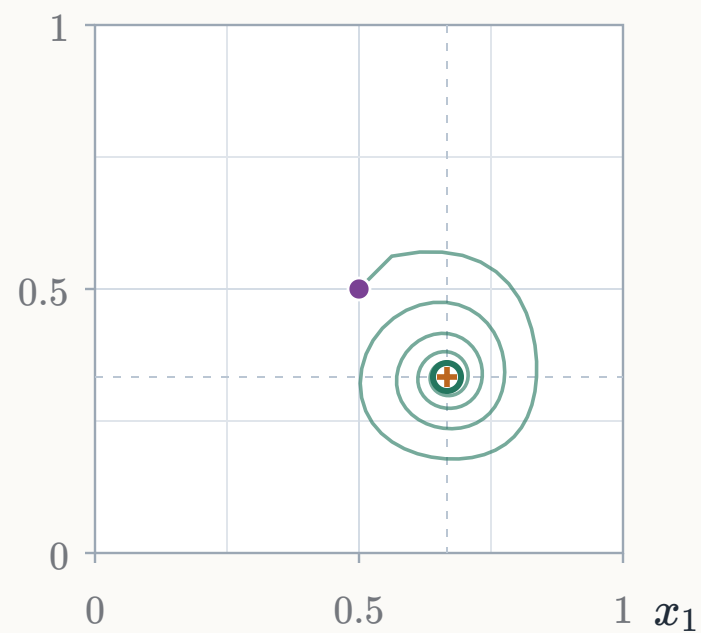
Constant $O_T(1)$ regret (2026, normal-form): [Liu et al.](#); [Abbadi et al.](#); [Soleymani & Piliouras](#).

Fast regret gives fast convergence of empirical play

$\text{Reg}_i^{(T)} \leq R(T) \ \forall i \implies \varepsilon_T \leq R(T)/T$. What about the last iterate?

– Optimistic learning dynamics

Optimistic MWU

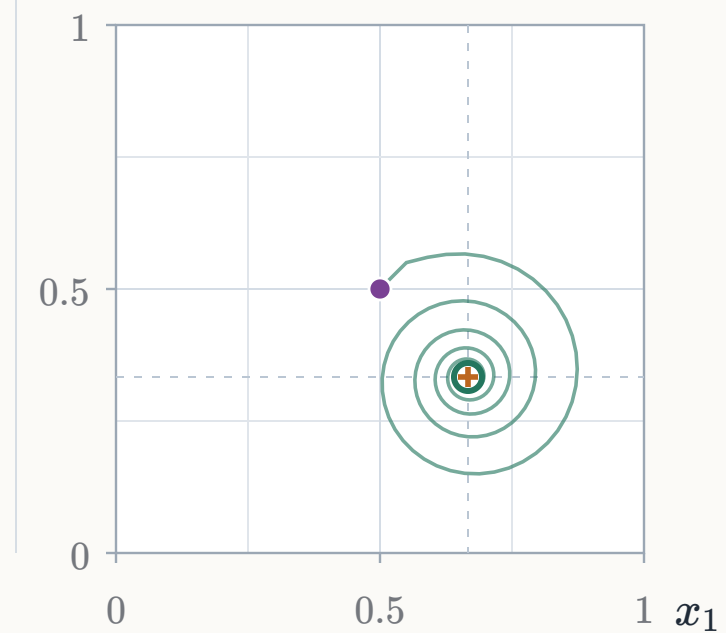


$$U = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

- Start
- Last iterate
- + Nash $(\frac{2}{3}, \frac{1}{3})$

Last-iterate gap
 2.7×10^{-7}

Optimistic OGD



$$\eta_{\text{MWU}} = 0.25$$

$$\eta_{\text{OGD}} = 0.1$$

- Start
- Last iterate
- + Nash $(\frac{2}{3}, \frac{1}{3})$

Last-iterate gap
 3.0×10^{-6}

(See [Supplementary S04](#).)

– Optimistic MWU can converge slowly

Ordinary MWU can cycle. Does optimism always fix the problem?

$$U_\delta = \begin{pmatrix} \frac{1}{2} & -\delta & \frac{1}{2} \\ 1 & & 0 \end{pmatrix}, \quad u_1 = \mathbf{x}^\top U_\delta \mathbf{y}, \quad u_2 = -u_1, \quad 0 < \delta < \frac{1}{2}$$

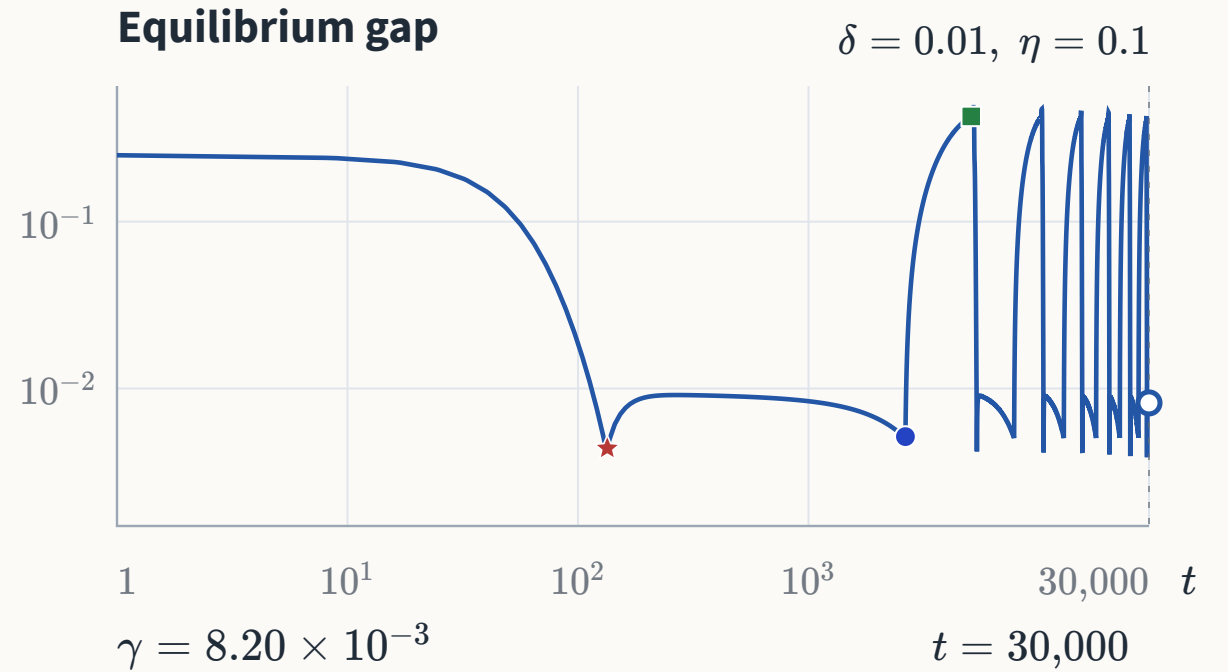
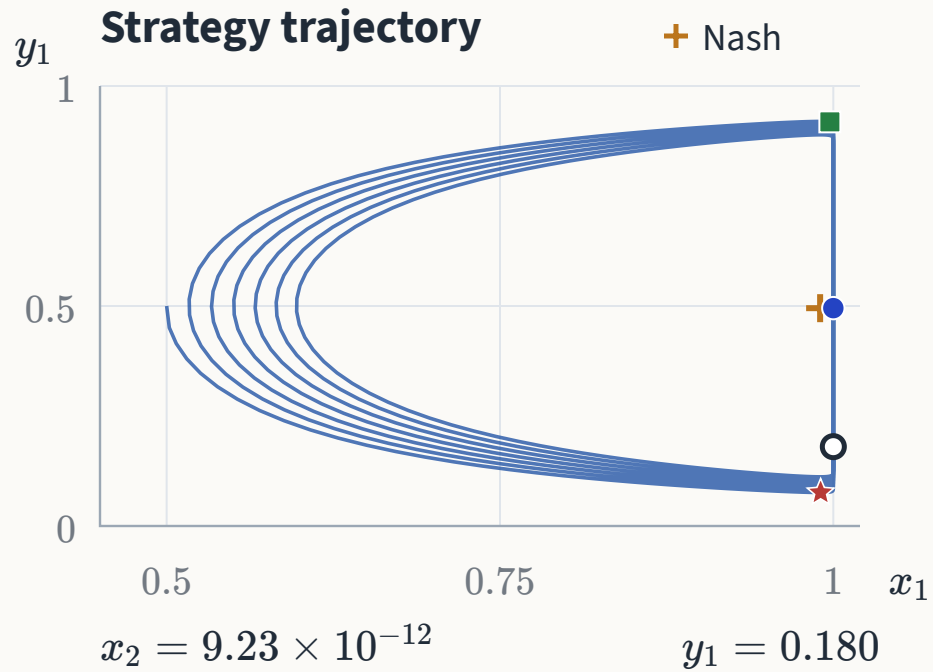
$$\mathbf{x}^* = \left(\frac{1}{1+\delta}, \frac{\delta}{1+\delta} \right), \quad \mathbf{y}^* = \left(\frac{1}{2(1+\delta)}, \frac{1+2\delta}{2(1+\delta)} \right)$$

An equilibrium close to the boundary

Optimistic MWU can retain an overwhelming preference for an action and overshoot the equilibrium.

→ [Cai, Farina, Grand-Clément, Kroer, Lee, Luo, Zheng \(2024\)](#).

– OMWU: trajectory and equilibrium gap



– The limit of optimistic MWU

No uniform last-iterate convergence rate

For fixed-stepsizes OMWU, no function of action counts and time alone gives a vanishing bound for every bounded-payoff game.

$$\gamma(\mathbf{x}^{(T)}, \mathbf{y}^{(T)}) \leq f(|A_1|, |A_2|, T), \quad f(|A_1|, |A_2|, T) \longrightarrow 0$$

Already impossible uniformly over 2×2 games with payoffs in $[0, 1]$.

Convergence in a fixed game is still possible

The rate must depend on the game. Optimistic OGD, by contrast, has a uniform $O(1/\sqrt{T})$ last-iterate guarantee.

[Cai et al., Fast Last-Iterate Convergence of Learning in Games Requires Forgetful Algorithms.](#)